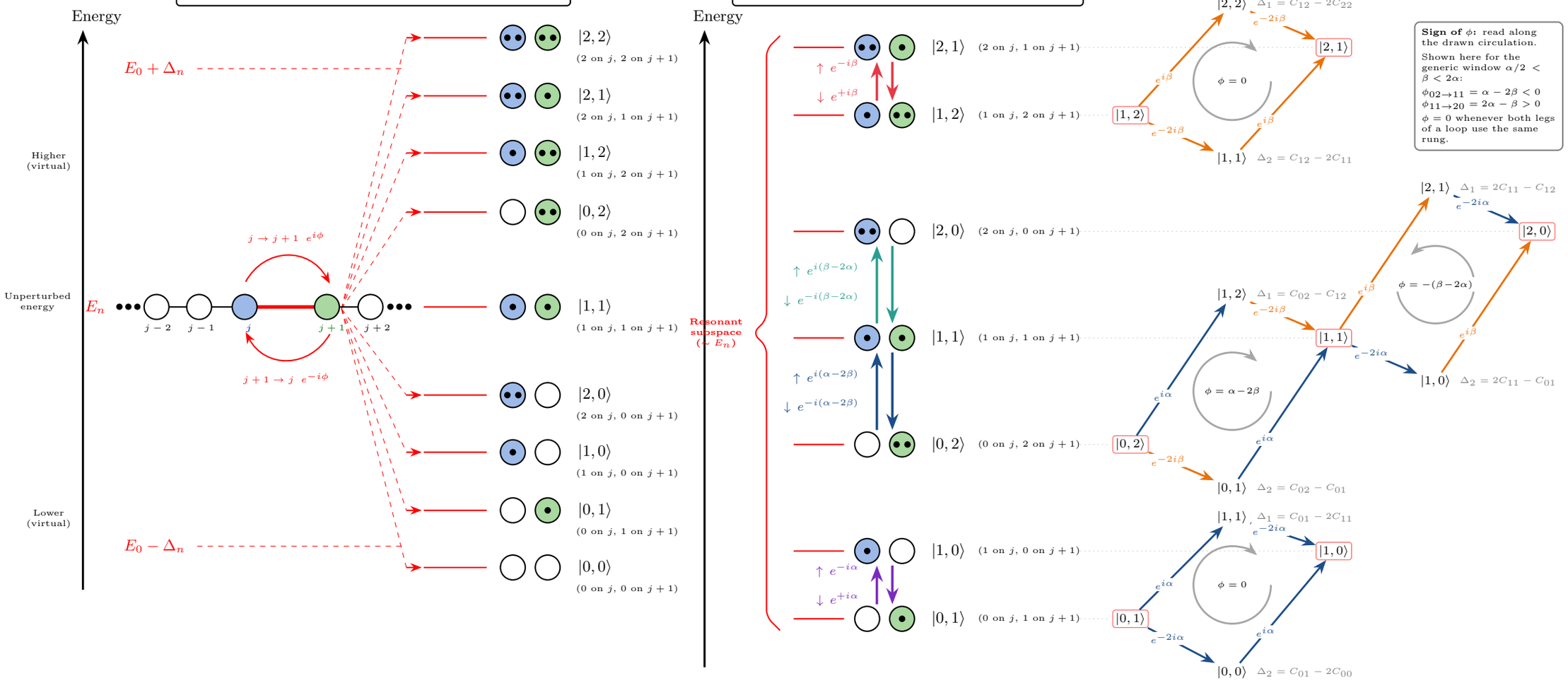
1D Rydberg lattice: active bond $(j, j+1)$

(A) Coupling of $|1,1\rangle$ to virtual (off–resonant) states

(B) Resonant subspace and its four transitions

(C) The two virtual paths behind each transition
navy = α rung $(0 \leftrightarrow 1)$, orange = β rung $(1 \leftrightarrow 2)$; boxed kets are the resonant levels of (B)



Sign of ϕ: read along the drawn circulation.
Shown here for the generic window $\alpha/2 < \beta < 2\alpha$:
$\phi_{02\to11} = \alpha - 2\beta < 0$
$\phi_{11\to20} = 2\alpha - \beta > 0$
$\phi = 0$ whenever both legs of a loop use the same rung.

(A) Where the virtual states come from. Each tweezer site carries an occupation $n = 0, 1, 2$; the Raman beams Ω_α $(0 \leftrightarrow 1)$ and Ω_β $(1 \leftrightarrow 2)$ move one excitation across the active bond $(j, j+1)$, imprinting $e^{i\phi}$ on a hop $j \to j+1$ and $e^{-i\phi}$ on the reverse hop.

Starting from $|1,1\rangle$ at energy E_n, a single hop reaches every other two–site configuration, but none of them is energy–matched: the interaction energies $C_{nn'}$ put them at $E_0 \pm \Delta_n$. Because $|\Delta_n| \gg \Omega$ these states are never populated — they are *virtual*, entered and left within one drive cycle.

Eliminating them to second order in Ω leaves an effective Hamiltonian acting only inside the resonant manifold, which is what panel (B) shows.

(B) What survives: the resonant subspace. The drive conserves the total occupation $N = n_j + n_{j+1}$ on the bond, so the resonant states split into blocks: $N = 3$ $\{|2,1\rangle, |1,2\rangle\}$, $N = 2$ $\{|2,0\rangle, |1,1\rangle, |0,2\rangle\}$ and $N = 1$ $\{|1,0\rangle, |0,1\rangle\}$, separated here by the wider gaps.

Only four second–order processes connect states inside a block; every other pair of levels is left untouched. Each pair carries two directed arrows: the upward transition acquires the phase printed beside it, the downward one its complex conjugate, as required by hermiticity.

The three $N = 2$ levels are degenerate only when $2C_{11} = C_{02}$. That resonance condition is what selects density–assisted hopping and makes the phase occupation dependent.

(C) How each transition is built. Every arrow of (B) is a two–photon process: it runs through one virtual state above the pair and one below. Boxed kets are the resonant levels of (B) and sit at exactly the same height (dotted guides); unboxed kets are virtual.

Each leg is labelled by the phase it imprints and coloured by the rung it uses — navy for α, orange for β. A leg acting on site j contributes $e^{+i\varphi}$, one acting on site $j+1$ contributes $e^{-2i\varphi}$, because the beam phases are $\varphi_j = \varphi$ and $\varphi_{j+1} = 2\varphi$.

The two paths of a process use the same two rungs in opposite order, so they carry the *same* total phase and differ only in their denominator, giving $A = \Omega\Omega' e^{i\phi}\left(\Delta_1^{-1} + \Delta_2^{-1}\right)$. The circular arrow marks that closed loop and ϕ is the statistical phase it encloses; it vanishes when both legs use the same rung. Since the two $N = 2$ processes share $|1,1\rangle$, they merge into a single chain.

Lattice links · Active bond $(j, j+1)$ · Site j · Site $j+1$ · Other sites · Resonant hopping (active–bond exchange) · Detuning from resonance (energy mismatch)